

**ADMINISTRATIVE INFORMATION****FMI No: 25410****FMI Title: Powerware UPS 9155-10GE – Inspection and Correction**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☒ Mandatory Safety	GEMS - AM:	Issue Date:
☐ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

- ☒ **Staged FMI** **Customer List** provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
- ☐ **Parts** and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
- ☐ **Parts Ordering:** GEMS - Americas, Contact your Region logistics coordinator
GEMS - Europe, Order FMI kits from GPO/EPL Buc
GEMS - Asia, Contact your region logistics coordinator
- ☐ **No Parts** are required to complete this FMI.
- ☐ **Batch with** _____
- To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).
- ☐ **Non-Staged** **No Customer List is provided.** Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Online Center 800-321-7937 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:

The purpose of this FMI is to inspect UPS Model 9155-10GE units installed in CT LightSpeed 1.X through 5.X for proper configuration and correct if not connected properly. The UPS units are those manufactured and shipped from January 2006 to January 2008.

Unit Tracked:**FMI Instructions Part No.: 5304992-1EN****Do NOT leave unsecured on site**

FMI Completion Certification Report

Mail To

GEMS - Americas	GEMS - Europe	
GE Medical Systems Attn: FMI Admin. W-523 P.O. Box 414 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

FMI No.: 25410

FMI Title: Powerware UPS 9155-10GE – Inspection and Correction

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer

Comments

Revision History

Revision	Date	Reason for Change
1	4/10/2008	Initial Release
2	5/16/2008	Changes made based on the field feedback

Effectivity

The effectivity list for this FMI was established by using all shipments of Uninterruptible Power Supply (UPS) (Model 9155-10GE) between January 2006 and January 2008. This list was obtained from Global OPSI for orders designated under catalog numbers B7999ZA and E4502KY. **Since the UPS is not a traceable part, Gantry model and serial numbers were used to create the effectivity list.** This list was considered to be complete at F4 release, so no further serials should be added.

Purpose

The purpose of this FMI is to inspect 2-Phase UPS Model 9155-10GE units installed between January 2006 and January 2008 in CT LightSpeed systems for proper configuration. Some of the units were shipped with a harness in the UPS not connected properly at the vendor's manufacturing facility. If not connected properly, the "Emergency OFF" feature is disabled. The UPS will not disconnect the batteries from the load and power will still be provided to the system.

iTRAK report numbers 13145242 and 13142781

Prerequisite FMIs

None

Furnished Materials

None.

Personnel Requirements

Travel Time: 1 hour

Labor Time: 1 hour

Procedure

Please see the following safety instructions before performing any work with the CT system.

Follow these general rules:

- All installed GE Healthcare systems require a LOTO compatible A1 Disconnect.
- Only qualified service personnel trained in the service and operation of this scanner, options, and accessories should perform any service on this equipment. Service personnel must observe and apply LOTO as defined by annual LOTO training.
- Equipment fuses, switches, and circuit breakers are for fire and equipment protection only. Do not rely on them to protect you against electrical shock or un-commanded equipment motion. This includes the Gantry Service Switch Panel!

Remove Power from the System

1. Prepare for shutdown of equipment. Notify affected personnel working in the area, via verbal communication, that LOTO is being performed.
2. Bring the system software down at the operator console.
3. After software shutdown is complete, wait two minutes and then place console power switch in the OFF position.
4. Turn off power at the system A1 Panel or Main Disconnect Panel. All personnel servicing the system must place their personal LOTO devices and tags on the A1 Panel or Main Disconnect Panel.
5. Verify that all energy has been dissipated as follows. Take the top and front cover off the PDU. Measure incoming system power at TS1 (for NGPDU) or Input Power Panel (for Compact PDU). Using a voltmeter or any other suitable meter, measure voltage from each phase to ground and verify zero energy.
6. Wait 5 minutes for any stored energy in capacitors to dissipate.
7. Shutdown the UPS unit.
8. Trip the battery breaker on the UPS unit.
9. Service/Repair CT system per service instructions.

Restore Power to the System

1. Notify affected personnel that equipment will be re-energized and LOTO devices are being removed.
2. All service personnel are to remove their personal LOTO lock and tag from A1 Panel or Main Disconnect Panel.
3. Re-apply power to the PDU Cabinet.
4. Reset the Gantry drives by pushing the reset button on any of the control panels.
5. Bring the system software back up.

1. Inspection of UPS Model 9155-10GE

Check to make sure that the system uses UPS Model 9155-10GE unit.

If the system does not use the UPS or uses an UPS of different Model number, this FMI is not applicable. Close the FMI as Dispatch Code 2, FMI not needed per instructions.

If the system uses UPS Model 9155-10GE, proceed with step 2.

To find the model number, remove the front panel of the top cabinet. To remove the top front panel, press and release the handle latch at the bottom center of the top cover and then lift the cover up and off the cabinet. The model number can be seen on the right side of the unit as shown in the picture below OR in the Rating Plate at the back of the unit.



Model Number is shown on the right side when the front cover is removed



Model Number is shown in the Rating Plate in the back of the unit.

It is located on the top left of the unit.

Fig. 1 Model Number of the unit

2. Inspection of UPS Model 9155-10GE for proper configuration

Remove front cover of the top UPS cabinet to inspect location X52 (see section 1 for instructions to remove the front cover). If harness cable from the top right of the unit is connected to X52 as shown in the Figures 2, the unit is configured correctly (either shipped correctly or previously completed).

Note: The harness cable could be ty-wrapped either through the top or bottom of the unit.



Fig. 2. Cable connection from top right of unit to X52.

Proceed to step 3, Rework Instructions if:

- A jumper is present at location X52 (as shown in Fig 3) and cable harness not connected
- No jumper in X52 and cable harness not connected
- Jumper in wrong location and cable harness not connected.

Proceed to step 5, Verification of 'Emergency Off' function if the harness is connected.

3. Rework Instructions

1. Perform Software shutdown.
2. Press any button on the front panel of UPS unit display to activate menu options.
3. Press the ↑ button on the front panel display and then press the → button to select TURN UPS ON/OFF menu.
4. Press the ↓ button to select the TURN UPS OFF option; press the ← button.
5. Confirm the selection. Press and hold the ← button for three seconds, until the UPS stops beeping; the input contactor should open.
6. Switch the UPS battery circuit breaker to the off position. The UPS disconnects from the batteries and is on logic power only.
7. Turn off system power by:
 - Depressing the 'STOP' button on the (A1) Main Disconnect **OR**
 - By turning Key Switch on the (A1) Main Disconnect to the 'OFF' position.

Follow proper safety procedures. Refer to the safety instructions under Procedures (page 2).

8. Verify the A1 panel shuts off immediately.
9. Verify the UPS shuts down.
10. Verify the UPS is not supplying any power to the system by observing the Gantry Displays. ALL GANTRY DISPLAY LIGHTS SHOULD BE OFF.
11. Remove Jumper Cable from connection point X52 on the UPS (Fig 3).



Fig. 3. Remove jumper cable in connector X52.

12. Gently pull cable from where it is protruding from the opening on the top right corner of the unit. Pull cable until it can be connected to location X52 (the spot where the jumper was connected).



Fig. 4, Cable protruding from the opening on top right corner of the unit.

13. Connect the pulled out cable to location X52.



Fig. 5. After the cable is connected to location X52

14. Position cable so that it can be secured to the bottom of the chassis with ty-wraps as shown in Fig 5. Secure cable with ty-wrap as shown.
15. Install the cover back on the UPS unit.

4. System Power Up

1. Verify the system is ready to receive power and all persons are clear of any potentially energized equipment.
2. Set the Manual By-Pass Switch to the “Service” position.
3. Verify the Load and Auxiliary circuit breakers on the UPS are in the “ON” position and the Battery breaker is “OFF”.
4. Verify the ‘LINE ON’ light on the Main Disconnect & UPS Control Panel (A1) is illuminated indicating mains power is available.
5. Verify the Key Switch in the A1 panel is in the “ON” position and none of the room SEO (System Emergency Off) buttons has been latched open.
6. Momentarily press the “START” button on the Main Disconnect & UPS Control Panel.
7. Verify ‘CONTACTOR ON’ and “UPS ON” lights on the Main Disconnect & UPS Control Panel are illuminated.
8. Re-set the UPS Battery breaker to the “ON” position.
9. Verify both the UPS and CT scanner begin to power up.
10. After approximately 30 seconds, the UPS should indicate electronic bypass mode. The UPS should display a flashing green “Inverter ON” indicator and a steady yellow “By-Pass ON” indicator.
11. When the yellow By-Pass indicator is ON steadily, switch the Manual By-Pass Switch to the “UPS” position.
12. After a few seconds the green “Inverter ON” indicator should switch to a steady “ON” and the Bypass indicator should turn off.
13. If the UPS fails to power up as indicated, refer to:

Powerware 9155-10GE UPS

10 kVA

User’s Guide

The User’s Guide can be found at:

http://data.supportcentral.ge.com/books/15768/B25579/doc_2972036.pdf

(Note: There is an underscore between doc and 2972036 in the above URL)

5. Verification of 'Emergency Off' function:

1. Depress any room System Emergency Off (SEO) button.
2. **Make sure that UPS applies no back up power to the CT system. The power to CT table/console and gantry should turn off and stay off.**
3. Follow the power up procedure indicated in Step 4 above to bring up system power.
4. Once the power is applied to the system and once the system is booted up, perform the following sanity scans to check system functionality:

50 axial scans at 120KV, 40 mA

The system should scan and image without any errors

6. FMI Closure

Close FMI per regional procedure. Remove FMI document from customer site.